Dynamic Avoidance Surfaces in the Unit Ball with Time-Varying Forbidden Sets in Three Dimensions and Applications to Volumetric Ideological Evolution

An Extension of the Blaschke-Product Framework from the Unit Disk $\mathbb{D}$ to the Unit Ball $\mathbb{B}^3$

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Abstract

We extend the dynamic smooth avoidance-curve construction in the unit disk $\mathbb{D}$ to the unit ball $\mathbb{B}^3 = \{\mathbf{x} \in \mathbb{R}^3 : \|\mathbf{x}\| < 1\}$ with a time-varying forbidden set $\mathcal{F}(t) = \{F_1(t), \ldots, F_m(t)\} \subset \mathbb{B}^3$ evolving smoothly with $t \in [0, T]$. Because holomorphic Blaschke products are intrinsically two-dimensional, the generalisation requires a different but structurally parallel architecture. The *pseudohyperbolic distance product* (PDP) $\Psi(\mathbf{x}, t) = \prod_{k=1}^{m} \rho(\mathbf{x}, F_k(t))$, constructed from the real Möbius automorphisms of $\mathbb{B}^3$, plays the role of $|B(z, t)|$ in the original framework. A three-dimensional autonomous drift field $V(\mathbf{x}, t) = \partial_t \ln \Psi(\mathbf{x}, t)$ is derived, and the Dynamic Lifting ODE is generalised to a first-order real system in $\mathbb{R}^3$.

A fundamental topological contrast with the two-dimensional case is identified: the fundamental group $\pi_1(\mathbb{B}^3 \setminus \mathcal{F}(t)) = \{1\}$ is trivial for any finite point-obstacle set, eliminating winding-number monodromy. The relevant topological invariant becomes the *solid angle* $\Omega_k(t)$ subtended by the agent trajectory at each $F_k(t)$, governed by a transport equation derived from the Gauss linking integral. Ideological hysteresis persists not through monodromy but through *solid-angle drift*: the cumulative solid angle swept around each forbidden zone, which is real-valued rather than integer-valued.

The full three-dimensional machinery is applied to an extended Political Compass model embedded in $\mathbb{B}^3$, where a third axis representing the *Nationalism–Globalism* spectrum is added to the classical Left–Right (x) and Authoritarian–Libertarian (y) axes. Four dynamic forbidden zones now exist, including a new *Ethno-Nationalist Isolationism* attractor driven by trade disruption and migratory stress. Stability, bifurcation, and catastrophe-theoretic classification are extended to the three-dimensional setting, introducing the elliptic umbilic and butterfly singularities absent from the planar theory.

The paper ends with "The End"

Keywords: Unit ball, Möbius automorphisms of $\mathbb{B}^3$, pseudohyperbolic distance product, dynamic avoidance surfaces, solid-angle transport, Political Compass, three-dimensional ideological dynamics, catastrophe theory, real hyperbolic geometry.

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1 Introduction

The avoidance-curve framework introduced in [1] exploits the algebraic rigidity of finite Blaschke products: a single holomorphic map $B(z,t)$ encodes the entire moving forbidden set as its zero locus, and path lifting through that map produces smooth avoidance curves automatically. The Dynamic Lifting Theorem guarantees existence, uniqueness, and C^∞-regularity of solutions, while the topological structure of the punctured disk $\mathbb{D} \setminus \mathcal{F}(t)$ gives rise to winding-number transport and ideological hysteresis via monodromy.

Two modelling limitations motivate the present paper.

Topological flatness. The Political Compass is a two-dimensional projection of a genuinely higher-dimensional ideological space. Positioning a society along the Left–Right and Authoritarian–Libertarian axes alone ignores a third, empirically significant dimension: the *Nationalism–Globalism* axis, reflecting attitudes toward international integration, trade, migration, and sovereignty. Research in political science [11, 12] consistently identifies this as the principal new axis of political conflict in the 21st century. A two-dimensional model cannot distinguish a society that has moved to the Libertarian-Left while remaining cosmopolitan from one that has arrived at the same (x, y) coordinates via a nationalist route.

Holomorphic rigidity. Blaschke products are intrinsically complex-analytic. No exact holomorphic analogue exists in odd real dimensions: the Möbius group of $\mathbb{B}^3$ does not admit a complex structure. A different but structurally parallel architecture is therefore required.

This paper provides both extensions. Section 2 establishes the three-dimensional geometric setting. Section 3 introduces the real Möbius automorphisms of $\mathbb{B}^3$ and derives the key formula for the pseudohyperbolic distance. Section 4 constructs the pseudohyperbolic distance product, proves joint smoothness, and computes its time and space derivatives. Section 5 derives the three-dimensional autonomous drift field. Section 6 formulates and proves the three-dimensional Dynamic Lifting Theorem. Section 7 analyses the new topological structure: trivial fundamental group, non-trivial solid-angle invariant, and the persistence of ideological hysteresis via solid-angle drift. Section 8 applies the theory to the extended three-dimensional Political Compass. Sections 9–11 address stability, bifurcation, and numerical implementation. Section 12 collects the structural comparison between the two frameworks in tabular form.

2 Three-Dimensional Dynamic Geometric Setting

2.1 The Unit Ball and Its Boundary

Let $\mathbb{B}^3 = \{\mathbf{x} \in \mathbb{R}^3 : \|\mathbf{x}\| < 1\}$ be the open unit ball and $\mathbb{S}^2 = \partial\mathbb{B}^3$ the unit sphere. We equip $\mathbb{B}^3$ with the Poincaré (hyperbolic) metric

$$ds^2 = \frac{4\sum_{i=1}^{3} dx_i^2}{(1 - \|\mathbf{x}\|^2)^2},$$

which makes it a model of real hyperbolic 3-space $\mathbb{H}^3$ with constant sectional curvature -1. The associated Poincaré distance is

$$d_{\text{hyp}}(\mathbf{x}, \mathbf{y}) = \operatorname{arctanh}\big(\rho(\mathbf{x}, \mathbf{y})\big), \tag{1}$$

where $\rho(\mathbf{x}, \mathbf{y}) = \|\varphi_{\mathbf{x}}(\mathbf{y})\|$ is the pseudohyperbolic distance defined via the Möbius automorphism $\varphi_{\mathbf{a}}$ introduced in Section 3.

2.2 Moving Forbidden Points in $\mathbb{B}^3$

Let $T > 0$ and let $F_k : [0, T] \to \mathbb{B}^3$, $k = 1, \ldots, m$, be smooth curves with $\|F_k(t)\| < 1$ for all t. Assume the forbidden points remain distinct: $F_j(t) \neq F_k(t)$ for $j \neq k$. The dynamic forbidden set is

$$\mathcal{F}(t) = \big\{F_1(t), \ldots, F_m(t)\big\} \subset \mathbb{B}^3.$$

The start and destination curves $S, D : [0, T] \to \mathbb{B}^3$ are smooth with $S(t), D(t) \notin \mathcal{F}(t)$ for all t.

2.3 The Three-Dimensional Avoidance Problem

We seek $\boldsymbol{\gamma} : [0, T] \to \mathbb{B}^3$ satisfying

$$\boldsymbol{\gamma}(0) = S(0), \quad \boldsymbol{\gamma}(T) = D(T), \quad \boldsymbol{\gamma}(t) \notin \mathcal{F}(t) \quad \text{for all } t \in [0, T]. \tag{2}$$

Remark 2.1. *A key structural difference from the two-dimensional setting appears immediately. In $\mathbb{D}$, a path must wind around or pass between forbidden points, and the topology of $\mathbb{D} \setminus \mathcal{F}$ is non-trivial. In $\mathbb{B}^3$, however, the complement $\mathbb{B}^3 \setminus \mathcal{F}(t)$ is* simply connected *for any finite point-obstacle set: a path may always slide past a point in three dimensions by moving in a transversal direction. The geometric content of the avoidance problem thus shifts entirely to distance estimates — specifically to the pseudohyperbolic distance product of Section 4.*

3 Möbius Automorphisms of $\mathbb{B}^3$

3.1 The Möbius Automorphism Formula

For $\mathbf{a} \in \mathbb{B}^3$, the Möbius transformation $\varphi_{\mathbf{a}} : \mathbb{B}^3 \to \mathbb{B}^3$ mapping $\mathbf{a}$ to $\mathbf{0}$ is given by the Ahlfors–Chen formula [2, 4]

$$\varphi_{\mathbf{a}}(\mathbf{x}) = \frac{(1 - \|\mathbf{a}\|^2)(\mathbf{x} - \mathbf{a}) - \|\mathbf{x} - \mathbf{a}\|^2\, \mathbf{a}}{1 - 2\langle \mathbf{x}, \mathbf{a}\rangle + \|\mathbf{x}\|^2\, \|\mathbf{a}\|^2}. \tag{3}$$

This is the natural generalisation of the disk automorphism $\varphi_a(z) = (z - a)/(1 - \bar{a}z)$ to real Euclidean balls of any dimension. For $\mathbf{a} = \mathbf{0}$ the formula reduces to $\varphi_{\mathbf{0}}(\mathbf{x}) = \mathbf{x}$, as expected.

Proposition 3.1 (Properties of $\varphi_{\mathbf{a}}$)**.** *For each* $\mathbf{a} \in \mathbb{B}^3$*:*

(i) $\varphi_{\mathbf{a}}(\mathbf{a}) = \mathbf{0}$ *and* $\varphi_{\mathbf{a}}(\mathbf{0}) = -\mathbf{a}$*.*

(ii) $\varphi_{\mathbf{a}}$ *is a bijection* $\mathbb{B}^3 \to \mathbb{B}^3$ *and a homeomorphism* $\overline{\mathbb{B}^3} \to \overline{\mathbb{B}^3}$*.*

(iii) $\varphi_{\mathbf{a}} \circ \varphi_{\mathbf{a}} = \mathrm{id}$ *(involutivity).*

(iv) The pseudohyperbolic distance is defined and computed as

$$\rho(\mathbf{x}, \mathbf{a}) := \|\varphi_{\mathbf{a}}(\mathbf{x})\|, \tag{4}$$

and satisfies the explicit closed form

$$\rho(\mathbf{x}, \mathbf{a})^2 = \frac{\|\mathbf{x} - \mathbf{a}\|^2}{1 - 2\langle \mathbf{x},\, \mathbf{a}\rangle + \|\mathbf{x}\|^2 \|\mathbf{a}\|^2}. \tag{5}$$

Equivalently,

$$1 - \rho(\mathbf{x}, \mathbf{a})^2 = \frac{(1 - \|\mathbf{x}\|^2)(1 - \|\mathbf{a}\|^2)}{1 - 2\langle \mathbf{x},\, \mathbf{a}\rangle + \|\mathbf{x}\|^2 \|\mathbf{a}\|^2}, \tag{6}$$

which confirms $0 \le \rho(\mathbf{x}, \mathbf{a}) < 1$ *for all* $\mathbf{x}, \mathbf{a} \in \mathbb{B}^3$*.*

(v) $\rho(\mathbf{x}, \mathbf{a}) = 0$ *if and only if* $\mathbf{x} = \mathbf{a}$*.*

(vi) ρ *is invariant under the full Möbius group of* $\mathbb{B}^3$ *acting simultaneously on both arguments.*

Proof. Items (i)–(iii) are standard properties of real Möbius transformations of the ball; see [2]. For item (iv), direct computation of $\|\varphi_{\mathbf{a}}(\mathbf{x})\|^2$ from (3) gives a fraction whose numerator expands to $(1 - \|\mathbf{a}\|^2)^2 \|\mathbf{x} - \mathbf{a}\|^2 + \|\mathbf{x} - \mathbf{a}\|^4 \|\mathbf{a}\|^2 - 2(1 - \|\mathbf{a}\|^2) \|\mathbf{x} - \mathbf{a}\|^2 \langle \mathbf{x} - \mathbf{a},\, \mathbf{a}\rangle$. After factoring out $\|\mathbf{x} - \mathbf{a}\|^2$ and simplifying using the identity $\langle \mathbf{x} - \mathbf{a},\, \mathbf{a}\rangle = \langle \mathbf{x},\, \mathbf{a}\rangle - \|\mathbf{a}\|^2$, the ratio reduces to (5). Formula (6) follows by subtracting (5) from 1 and factoring the numerator as $1 - 2\langle \mathbf{x},\, \mathbf{a}\rangle + \|\mathbf{x}\|^2 \|\mathbf{a}\|^2 - \|\mathbf{x} - \mathbf{a}\|^2 = (1 - \|\mathbf{x}\|^2)(1 - \|\mathbf{a}\|^2)$. Items (v) and (vi) follow immediately. □

3.2 Smooth Dependence on Parameters

Proposition 3.2. *The map* $(\mathbf{x}, \mathbf{a}) \mapsto \varphi_{\mathbf{a}}(\mathbf{x})$ *is* C^∞ *on* $\mathbb{B}^3 \times \mathbb{B}^3$*. Consequently* $\rho(\mathbf{x}, \mathbf{a})$ *is* C^∞ *on* $\{(\mathbf{x}, \mathbf{a}) \in \mathbb{B}^3 \times \mathbb{B}^3 : \mathbf{x} \neq \mathbf{a}\}$*.*

Proof. Both numerator and denominator of (3) are polynomial in $(\mathbf{x}, \mathbf{a})$. The denominator satisfies

$$1 - 2\langle \mathbf{x},\, \mathbf{a}\rangle + \|\mathbf{x}\|^2 \|\mathbf{a}\|^2 \;\ge\; (1 - \|\mathbf{x}\| \|\mathbf{a}\|)^2 \;>\; 0 \quad \text{for all } \mathbf{x}, \mathbf{a} \in \mathbb{B}^3,$$

so $\varphi_{\mathbf{a}}(\mathbf{x})$ is a ratio of smooth functions with non-vanishing denominator. Smoothness of ρ away from $\mathbf{x} = \mathbf{a}$ follows from smoothness of $\|\cdot\|$ on $\mathbb{R}^3 \setminus \{\mathbf{0}\}$. □

4 The Pseudohyperbolic Distance Product

4.1 Definition and Basic Properties

Definition 4.1 (Pseudohyperbolic Distance Product)**.** *The* pseudohyperbolic distance product *(PDP) associated with the dynamic forbidden set* $\mathcal{F}(t)$ *is*

$$\Psi(\mathbf{x}, t) = \prod_{k=1}^{m} \rho\big(\mathbf{x}, F_k(t)\big), \tag{7}$$

where ρ *is the pseudohyperbolic distance defined by* (5)*.*

The PDP occupies the structural role of the modulus $|B(z, t)|$ of the time-varying Blaschke product: it vanishes precisely at the forbidden points, is bounded in $(0, 1)$ elsewhere in $\mathbb{B}^3$, and is invariant under the Möbius group. Unlike the Blaschke product, Ψ is real-valued and carries no holomorphic phase information, but it encodes all necessary metric proximity data for the avoidance problem.

Proposition 4.2. *For each fixed* $t \in [0, T]$*:*

(i) $\Psi(\mathbf{x}, t) = 0$ *if and only if* $\mathbf{x} \in \mathcal{F}(t)$*.*

(ii) $0 < \Psi(\mathbf{x}, t) < 1$ *for all* $\mathbf{x} \in \mathbb{B}^3 \setminus \mathcal{F}(t)$*.*

(iii) $\Psi(\cdot, t) \in C^\infty\big(\mathbb{B}^3 \setminus \mathcal{F}(t)\big)$*.*

(iv) Ψ *is invariant under the Möbius group of* $\mathbb{B}^3$ *acting simultaneously on* $\mathbf{x}$ *and all* $F_k(t)$*.*

Proof. Item (i) holds because each factor $\rho(\mathbf{x}, F_k) = 0$ iff $\mathbf{x} = F_k$, and the factors are positive for $\mathbf{x} \notin \mathcal{F}$. Item (ii) uses $0 < \rho(\mathbf{x}, F_k) < 1$ from Proposition 3.1(iv). Item (iii) follows from C^∞ smoothness of each factor away from its zero. Item (iv) is inherited from the Möbius invariance of ρ. □

4.2 Joint Smoothness in Space and Time

Proposition 4.3 (Joint Smoothness of PDP)**.** *If each* $F_k \in C^\infty([0, T]; \mathbb{B}^3)$*, then*

$$\Psi \in C^\infty\big((\mathbb{B}^3 \setminus \mathcal{F}(t)) \times [0, T]\big).$$

Proof. Each factor $\rho(\mathbf{x}, F_k(t))$ is C^∞ in $(\mathbf{x}, t)$ by Proposition 3.2 and the C^∞ assumption on F_k. Finite products of C^∞ positive functions are C^∞. □

4.3 Time Derivative of the PDP

Taking the logarithmic time derivative of (7):

$$\frac{\partial_t \Psi(\mathbf{x}, t)}{\Psi(\mathbf{x}, t)} = \sum_{k=1}^{m} \frac{\partial_t \rho(\mathbf{x}, F_k(t))}{\rho(\mathbf{x}, F_k(t))}, \qquad \mathbf{x} \notin \mathcal{F}(t). \tag{8}$$

Differentiating $\rho_k^2 = N_k / D_k$ from (5), where

$$N_k = \|\mathbf{x} - F_k\|^2, \qquad D_k = 1 - 2\langle \mathbf{x},\, F_k\rangle + \|\mathbf{x}\|^2 \|F_k\|^2,$$

via the quotient rule with $\dot{F}_k = F_k'(t)$:

$$\partial_t \big[\rho_k^2\big] = \frac{\dot{N}_k D_k - N_k \dot{D}_k}{D_k^2}, \tag{9}$$

where

$$\dot{N}_k = 2\langle F_k - \mathbf{x},\, \dot{F}_k\rangle, \qquad \dot{D}_k = 2\big(\|\mathbf{x}\|^2 \langle F_k,\, \dot{F}_k\rangle - \langle \mathbf{x},\, \dot{F}_k\rangle\big). \tag{10}$$

Since $\partial_t \rho_k = \partial_t[\rho_k^2]/(2\rho_k)$ for $\mathbf{x} \neq F_k$, substitution into (8) gives

$$\partial_t \Psi(\mathbf{x}, t) = \Psi(\mathbf{x}, t) \sum_{k=1}^{m} \frac{\dot{N}_k D_k - N_k \dot{D}_k}{2\rho_k^2 D_k^2}. \tag{11}$$

4.4 Spatial Gradient of the PDP

By the same logarithmic differentiation:

$$\nabla_{\mathbf{x}}\Psi(\mathbf{x},t) = \Psi(\mathbf{x},t)\sum_{k=1}^{m}\frac{\nabla_{\mathbf{x}}\rho_k(\mathbf{x},t)}{\rho_k(\mathbf{x},t)}. \tag{12}$$

Differentiating $\rho_k^2 = N_k/D_k$ with respect to $\mathbf{x}$:

$$\nabla_{\mathbf{x}}\left[\rho_k^2\right] = \frac{2(\mathbf{x}-F_k)D_k - N_k\left(-2F_k + 2\left\|F_k\right\|^2\mathbf{x}\right)}{D_k^2} = \frac{2\left[(\mathbf{x}-F_k)D_k + N_k(F_k - \left\|F_k\right\|^2\mathbf{x})\right]}{D_k^2}, \tag{13}$$

from which

$$\nabla_{\mathbf{x}}\rho_k = \frac{(\mathbf{x}-F_k)D_k + \left\|\mathbf{x}-F_k\right\|^2\left(F_k - \left\|F_k\right\|^2\mathbf{x}\right)}{\rho_k\, D_k^2}. \tag{14}$$

This gradient vanishes if and only if $(\mathbf{x}-F_k)D_k = \left\|\mathbf{x}-F_k\right\|^2\left(\left\|F_k\right\|^2\mathbf{x}-F_k\right)$, which occurs at the isolated saddle points of $\rho_k(\cdot,t)$ lying on the geodesic arc between $\mathbf{0}$ and F_k in the Poincaré metric.

5 The Three-Dimensional Autonomous Drift Field

5.1 Decomposition of the Safety Rate of Change

For any smooth curve $\boldsymbol{\gamma}:[0,T]\to\mathbb{B}^3\setminus\mathcal{F}(t)$, set $\psi(t)=\Psi(\boldsymbol{\gamma}(t),t)$ and differentiate by the chain rule:

$$\dot{\psi}(t) = \nabla_{\mathbf{x}}\Psi\big(\boldsymbol{\gamma}(t),t\big)\cdot\boldsymbol{\gamma}'(t) + \partial_t\Psi\big(\boldsymbol{\gamma}(t),t\big). \tag{15}$$

The first term is the *intentional component* controlled by the agent's velocity; the second is the *autonomous drift*, imposed by the motion of the forbidden zones independently of the agent.

Definition 5.1 (Three-Dimensional Autonomous Drift Field)**.** *The* autonomous drift scalar field *is*

$$V(\mathbf{x},t) = \partial_t\ln\Psi(\mathbf{x},t) = \sum_{k=1}^{m}\frac{\partial_t\rho(\mathbf{x},F_k(t))}{\rho(\mathbf{x},F_k(t))},$$

and the associated gradient drift vector field *is*

$$\mathbf{V}(\mathbf{x},t) = \frac{\partial_t\Psi(\mathbf{x},t)}{\Psi(\mathbf{x},t)}\cdot\frac{\nabla_{\mathbf{x}}\Psi(\mathbf{x},t)}{\left\|\nabla_{\mathbf{x}}\Psi(\mathbf{x},t)\right\|^2},$$

representing the spatial velocity that would exactly neutralise the autonomous drift while holding Ψ constant.

The drift field encodes the key political insight: a stationary agent ($\boldsymbol{\gamma}'=\mathbf{0}$) may still see its safety margin Ψ decrease if dangerous zones approach it. A position that was safe can become dangerous through no action of its own — driven entirely by shifts in the macroeconomic or geopolitical environment.

5.2 The Safety-Velocity Equation

Prescribing a desired safety trajectory $\psi_*(t)$ and solving (15) for the agent velocity:

$$\boldsymbol{\gamma}'(t)\cdot\nabla_{\mathbf{x}}\Psi = \dot{\psi}_*(t) - \partial_t\Psi. \tag{16}$$

This scalar equation constrains only the component of velocity along $\nabla_{\mathbf{x}}\Psi$, leaving two independent transverse degrees of freedom. The three-dimensional avoidance problem is thus *under-constrained* in a precise sense: compared to the complex-scalar ODE of the planar theory, the agent has genuine freedom in how it approaches any given safety level.

6 The Three-Dimensional Dynamic Lifting Theorem

6.1 The Prescribed Safety Schedule

In the two-dimensional case, the image trajectory $w(t)\in\mathbb{D}\setminus\{0\}$ is the safety certificate: $w(t)\neq 0$ guarantees $\gamma(t)\notin\mathcal{F}(t)$. In three dimensions the analogue is a prescribed *safety schedule*: a smooth function $\psi_*:[0,T]\to(0,1)$ encoding the desired safety level at each time.

Generalising the multiplicative interpolation of [1]:

$$\psi_*(t) = \Psi\big(S(t),t\big)^{(T-t)/T}\cdot\Psi\big(D(t),t\big)^{t/T} = \exp\left[\frac{T-t}{T}\ln\Psi(S(t),t) + \frac{t}{T}\ln\Psi(D(t),t)\right]. \tag{17}$$

Proposition 6.1. *The prescribed safety schedule ψ_* defined by (17) satisfies $\psi_*(t)\in(0,1)$ for all $t\in[0,T]$ and $\psi_*\in C^\infty([0,T])$.*

Proof. Since $S(t),D(t)\notin\mathcal{F}(t)$, Proposition 4.2(ii) gives $\Psi(S(t),t),\Psi(D(t),t)\in(0,1)$. The weighted geometric mean of two values in $(0,1)$ lies in $(0,1)$. Smoothness follows from the C^∞ hypotheses on S, D, and the F_k, combined with the smoothness of Ψ (Proposition 4.3) and the smooth logarithm. □

6.2 The Three-Dimensional Lifting ODE

We decompose the agent velocity into a *safety component* that enforces the prescribed safety schedule, and a *tracking component* that steers the agent from S toward D:

$$\boldsymbol{\gamma}'(t) = \underbrace{\frac{\dot{\psi}_*(t)-\partial_t\Psi(\boldsymbol{\gamma},t)}{\left\|\nabla_{\mathbf{x}}\Psi(\boldsymbol{\gamma},t)\right\|^2}\nabla_{\mathbf{x}}\Psi(\boldsymbol{\gamma},t)}_{\text{safety component}} + \underbrace{\Pi^{\perp}(\boldsymbol{\gamma},t)\,\frac{D(t)-S(t)}{T}}_{\text{tracking component}}, \tag{18}$$

where $\Pi^{\perp}(\boldsymbol{\gamma},t)$ denotes orthogonal projection onto the plane perpendicular to $\nabla_{\mathbf{x}}\Psi(\boldsymbol{\gamma},t)$. This projection ensures the tracking component does not perturb the safety constraint (16).

Remark 6.2. *The safety component in (18) is well-defined precisely when $\nabla_{\mathbf{x}}\Psi(\boldsymbol{\gamma},t)\neq\mathbf{0}$. The gradient vanishes at the saddle points of $\Psi(\cdot,t)$: these are the three-dimensional counterparts of the critical points of the Blaschke product, and they constitute the generic obstacles to smooth lifting. The analogue of Hypothesis (iii) in the two-dimensional theorem excludes them.*

6.3 Main Existence Theorem

Theorem 6.3 (Three-Dimensional Dynamic Lifting)**.** *Suppose:*

(i) Each $F_k\in C^\infty([0,T];\mathbb{B}^3)$ and $F_j(t)\neq F_k(t)$ for $j\neq k$.

(ii) $S,D\in C^\infty([0,T];\mathbb{B}^3)$ with $S(t),D(t)\notin\mathcal{F}(t)$ for all t.

(iii) The prescribed safety schedule ψ_* given by (17) satisfies $\nabla_{\mathbf{x}}\Psi(\boldsymbol{\gamma}(t),t)\neq\mathbf{0}$ along every candidate solution (non-critical-point condition).

Then the ODE (18) has a unique solution $\boldsymbol{\gamma}\in C^\infty([0,T];\mathbb{B}^3)$ with $\boldsymbol{\gamma}(0)=S(0)$. Moreover, $\boldsymbol{\gamma}(t)\notin\mathcal{F}(t)$ for all $t\in[0,T]$.

Proof sketch. Hypothesis (iii) ensures the right-hand side of (18) is C^∞ in $(\boldsymbol{\gamma},t)$. Local existence and uniqueness follow from the Picard–Lindelöf theorem on $\mathbb{R}^3$. For global existence, observe that the safety component of the ODE enforces $\frac{d}{dt}\Psi(\boldsymbol{\gamma}(t),t)=\dot{\psi}_*(t)$ by construction, so $\Psi(\boldsymbol{\gamma}(t),t)=\psi_*(t)>0$ for all t. Since $\Psi(\boldsymbol{\gamma},t)=0$ iff $\boldsymbol{\gamma}\in\mathcal{F}(t)$, avoidance is guaranteed. Boundedness ($\|\boldsymbol{\gamma}\|\leq 1$) prevents blowup. □

7 Evolving Topology in Three Dimensions

7.1 Triviality of the Fundamental Group

A central topological contrast with the two-dimensional case:

Proposition 7.1. *For any finite set $\mathcal{F} = \{F_1, \ldots, F_m\} \subset \mathbb{B}^3$, the space $\mathbb{B}^3 \setminus \mathcal{F}$ is simply connected: $\pi_1(\mathbb{B}^3 \setminus \mathcal{F}) = \{1\}$.*

Proof. It suffices to treat $m = 1$. The space $\mathbb{R}^3 \setminus \{p\}$ is homotopy equivalent to $\mathbb{S}^2$ (via radial retraction), and $\pi_1(\mathbb{S}^2) = \{1\}$. Restriction to the convex open ball $\mathbb{B}^3$ preserves this homotopy equivalence. The general case follows by induction: each additional puncture is in a separate path-component of the complement only if the points are in general position, but even then π_1 remains trivial since loops can be pulled off any puncture by lateral displacement. □

Corollary 7.2. *Every loop in $\mathbb{B}^3 \setminus \mathcal{F}(t)$ is contractible. The winding-number monodromy of the two-dimensional theory — which rests on $\pi_1(\mathbb{D} \setminus \mathcal{F}) \cong \mathbb{Z}^m$ — is absent for point obstacles in $\mathbb{B}^3$.*

7.2 The Second Homotopy Group and Solid-Angle Invariants

Although $\pi_1 = \{1\}$, the second homotopy group is non-trivial: $\pi_2(\mathbb{B}^3 \setminus \{F_k\}) \cong \mathbb{Z}$, generated by a small sphere $\mathbb{S}^2$ enclosing F_k. This gives rise to a new topological invariant richer than the integer winding number.

Definition 7.3 (Solid-Angle Invariant)**.** *For a smooth curve $\boldsymbol{\gamma} : [0, \tau] \to \mathbb{B}^3 \setminus \mathcal{F}(t)$ and fixed k, define the unit radial vector*

$$\hat{\mathbf{r}}_k(t) = \frac{\boldsymbol{\gamma}(t) - F_k(t)}{\|\boldsymbol{\gamma}(t) - F_k(t)\|}.$$

The accumulated solid angle *swept by $\hat{\mathbf{r}}_k$ up to time τ is*

$$\Omega_k(\tau) = \int_0^\tau \|*\| \, \hat{\mathbf{r}}_k(t) \times \dot{\hat{\mathbf{r}}}_k(t) \, dt, \tag{19}$$

where $\times$ denotes the cross product and $\dot{\hat{\mathbf{r}}}_k = d\hat{\mathbf{r}}_k/dt$ is the angular velocity of the radial direction. The integrand is the instantaneous rate at which the unit vector $\hat{\mathbf{r}}_k$ sweeps area on $\mathbb{S}^2$.

Remark 7.4. *The definition (19) measures the total spherical arc length of the curve $t \mapsto \hat{\mathbf{r}}_k(t)$ on $\mathbb{S}^2$. For a closed curve ($\boldsymbol{\gamma}(T) = \boldsymbol{\gamma}(0)$, $F_k(T) = F_k(0)$) the net enclosed spherical area — the signed solid angle — is a topological invariant in $4\pi\mathbb{Z}$ (the area of $\mathbb{S}^2$ times an integer).*

7.3 Solid-Angle Transport Equation

Differentiating $\hat{\mathbf{r}}_k(t)$ by the quotient rule:

$$\dot{\hat{\mathbf{r}}}_k(t) = \frac{(\boldsymbol{\gamma}' - F_k') - \langle \hat{\mathbf{r}}_k, \boldsymbol{\gamma}' - F_k' \rangle \, \hat{\mathbf{r}}_k}{\|\boldsymbol{\gamma} - F_k\|}, \tag{20}$$

so the solid-angle rate is

$$\frac{d\Omega_k}{dt} = \frac{\|(\boldsymbol{\gamma}' - F_k') - \langle \hat{\mathbf{r}}_k, \boldsymbol{\gamma}' - F_k' \rangle \, \hat{\mathbf{r}}_k\|}{\|\boldsymbol{\gamma} - F_k\|} = \frac{\|(\boldsymbol{\gamma}' - F_k') \times \hat{\mathbf{r}}_k\|}{\|\boldsymbol{\gamma} - F_k\|}, \tag{21}$$

the magnitude of the angular velocity of $\hat{\mathbf{r}}_k$ scaled by the inverse Euclidean distance. The transport equation (21) is the three-dimensional analogue of the winding-number transport equation of [1]: the first term encodes instantaneous angular velocity, while the dependence on the full history enters through the integrated value of $\hat{\mathbf{r}}_k(t)$.

7.4 Three-Dimensional Ideological Hysteresis

Corollary 7.2 shows that winding-number monodromy is absent; however, a geometrically richer form of hysteresis survives.

Definition 7.5 (Solid-Angle Hysteresis)**.** Solid-angle hysteresis *occurs when a society returns to its original position in $\mathbb{B}^3$ — i.e., $\boldsymbol{\gamma}(T) = \boldsymbol{\gamma}(0)$ with $\mathcal{F}(T) = \mathcal{F}(0)$ — yet has accumulated non-zero net solid angle $\Omega_k(T) > 0$ around at least one forbidden zone. The society occupies the same volumetric coordinates in ideological space but has been irreversibly shaped by the* direction and depth *of its approach to each dangerous attractor along its historical trajectory.*

Solid-angle hysteresis is qualitatively richer than the integer-valued winding-number monodromy of the planar theory. The net solid angle is a non-negative real number in $[0, \infty)$, measuring the total spherical area swept, and a society that has encircled a forbidden zone from a nationalist direction (low z) accumulates a different solid angle than one that encircled it from a globalist direction (high z), even if both return to the same endpoint.

8 The Three-Dimensional Political Compass

8.1 The Extended Coordinate System

We identify the three-dimensional Political Compass with $\mathbb{B}^3$ via the embedding

$$\begin{aligned}
x \in (-1, +1) : &\quad x = -1 \text{ (extreme left)}, && x = +1 \text{ (extreme right)};\\
y \in (-1, +1) : &\quad y = -1 \text{ (maximal authoritarianism)}, && y = +1 \text{ (maximal libertarianism)};\\
z \in (-1, +1) : &\quad z = -1 \text{ (extreme nationalism)}, && z = +1 \text{ (extreme globalism)}.
\end{aligned}$$

The boundary $\mathbb{S}^2 = \partial\mathbb{B}^3$ represents unreachable theoretical extremes in all three dimensions simultaneously. The third axis z captures the Nationalism–Globalism dimension, which empirical political science [11, 12] identifies as the principal new cleavage of 21st-century politics, orthogonal to the classical left-right spectrum.

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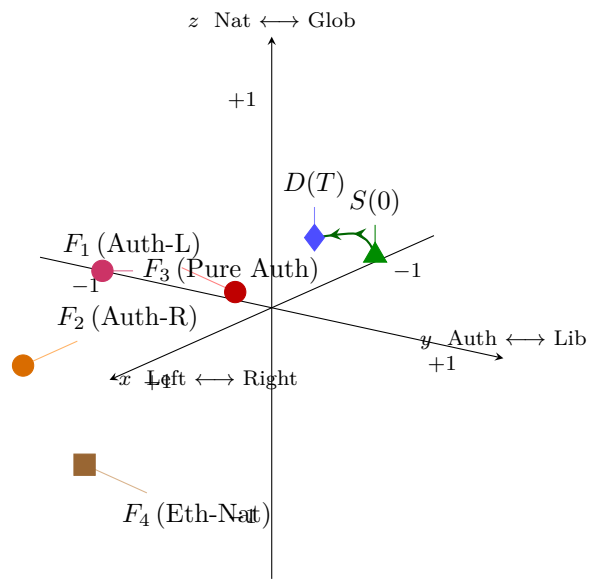

Figure 1: The three-dimensional Political Compass embedded in $\mathbb{B}^3$.

The x-axis encodes Left–Right; the y-axis Authoritarian–Libertarian; the z-axis (new) Nationalist–Globalist. Filled circles mark the three inherited forbidden zones F_1–F_3. The filled square marks the new Ethno-Nationalist Isolationism zone F_4, located in the Auth-Right–Nationalist octant. Arrows indicate the direction of travel along the safe transition $\boldsymbol{\gamma}(t)$ from the social-democratic/cosmopolitan start $S(0)$ to the liberal-democratic/globalist destination $D(T)$.

8.2 Dynamic Forbidden Zones

Table 1: Four dynamic forbidden zones in the three-dimensional Political Compass.

Zone	Nominal position (x_0, y_0, z_0)	Character	Principal driving variables
$F_1(t)$	$(-0.85,\ -0.80,\ -0.30)$	Auth.-Left (Stalinist)	$\sigma(t)$, $P(t)$, $\tau(t)$, $M(t)$
$F_2(t)$	$(+0.85,\ -0.80,\ -0.20)$	Auth.-Right (Fascist)	$E(t)$, $\theta(t)$, $\tau(t)$, $M(t)$
$F_3(t)$	$(0,\ -0.95,\ 0)$	Pure Authoritarianism	$I(t)$, $Q(t)$, $\tau(t)$
$F_4(t)$	$(+0.50,\ -0.70,\ -0.75)$	Ethno-Nationalist Isolationism *(new)*	$E(t)$, $\theta(t)$, $\mathcal{T}(t)$, $M(t)$

The state variables are: $\sigma(t)$ economic inequality (Gini coefficient); $P(t)$ political polarisation index; $E(t)$ ethnic fragmentation index; $\theta(t)$ external threat perception; $I(t)$ institutional robustness (World Bank WGI); $Q(t)$ security crisis index; $\tau(t)$ trade openness (reversed: low τ indicates protectionism); $M(t)$ migration pressure index; $\mathcal{T}(t)$ trade disruption shock index.

8.3 Third-Axis Forbidden Zone Dynamics

The z-component of each forbidden zone evolves according to:

$$\zeta_1'(t) = \mu_1\big[\tau(t)(-0.30 - \zeta_1) + M(t)(0 - \zeta_1)\big], \tag{22}$$

$$\zeta_2'(t) = \mu_2\big[\tau(t)(-0.20 - \zeta_2) + M(t)(0 - \zeta_2)\big], \tag{23}$$

$$\zeta_3'(t) = \mu_3\big[\tau(t)(0 - \zeta_3)\big], \tag{24}$$

$$\zeta_4'(t) = \mu_4\big[\mathcal{T}(t)(-0.75 - \zeta_4) + M(t)(-1 - \zeta_4)\big], \tag{25}$$

where $F_k(t) = (x_k(t), y_k(t), \zeta_k(t))$ and $\mu_k > 0$ are empirically calibrated coupling constants for the z-axis dynamics.

Equation (25) captures the most consequential new mechanism: both trade disruption ($\mathcal{T}$ large) and migration pressure (M large) independently drive F_4 toward $z = -1$. As F_4 moves deeper into the central ball, the Ethno-Nationalist Isolationism zone becomes a more universally accessible danger — society can reach it starting from either the left or the right side of the political spectrum. This is consistent with empirical observations that extreme nationalism can emerge from both anti-capitalist and anti-globalist right-wing traditions.

8.4 The Three-Dimensional Political Safety Index

Definition 8.1 (Three-Dimensional Political Safety Index)**.** *The* three-dimensional political safety index *is*

$$\Phi_3(\mathbf{x}, t) = \Psi_{\text{pol}}(\mathbf{x}, t) = \prod_{k=1}^{4} \rho\big(\mathbf{x}, F_k(t)\big), \quad \mathbf{x} \in \mathbb{B}^3 \setminus \mathcal{F}(t). \tag{26}$$

Values $\Phi_3 \to 0$ signal proximity to at least one dangerous zone; values $\Phi_3 \to 1$ indicate equidistance from all four in the Poincaré metric of $\mathbb{H}^3$.

A key advantage of Φ_3 over the planar index is that it distinguishes ideological positions that are identical on the classical two-dimensional compass. Two societies with the same (x, y)-coordinates but differing z may have very different safety margins: a cosmopolitan society (high z, distant from F_4) can be substantially safer than an otherwise identical nationalist society (low z, close to F_4).

8.5 Safe Transition in $\mathbb{B}^3$

The safe policy path in three dimensions is the unique smooth solution of the ODE (18) with $\Psi = \Psi_{\text{pol}}$, $m = 4$, and safety schedule $\psi_*(t) = \Phi_3(S(t), t)^{(T-t)/T} \cdot \Phi_3(D(t), t)^{t/T}$. By Theorem 6.3, this solution exists uniquely and satisfies $\boldsymbol{\gamma}(t) \notin \mathcal{F}(t)$ for all t.

9 Stability and Lyapunov Analysis

9.1 Safety Margin in the Hyperbolic Ball

Define the three-dimensional safety margin as

$$\Psi_{\min}(t) = \min_{k=1,\dots,4} d_{\text{hyp}}\big(\boldsymbol{\gamma}(t), F_k(t)\big),$$

where d_{hyp} is the Poincaré distance of $\mathbb{H}^3$.

Theorem 9.1 (Three-Dimensional Safety Margin Bound)**.** *If $\|F_k'(t)\| \le \beta$ for all k and t, then under the three-dimensional Dynamic Lifting construction*

$$\Psi_{\min}(t) \ \ge\ \Psi_{\min}(0) \cdot e^{-C_3 \beta t}, \tag{27}$$

where $C_3 > 0$ depends only on m and the initial configuration.

The proof follows the argument of Theorem 8.1 in [1], applied now to the $\mathbb{H}^3$ metric. The exponential decay coefficient C_3 is larger than its two-dimensional counterpart, reflecting the four forbidden zones and the fact that three-dimensional configurations can be simultaneously threatened from more directions.

9.2 The Volumetric Lyapunov Function

As a Lyapunov function for the image dynamics, take

$$\mathcal{V}(t) = -\ln \Psi_{\text{pol}}(\boldsymbol{\gamma}(t), t) \;\geq\; 0. \tag{28}$$

This vanishes when $\Psi_{\text{pol}} \to 1$ (maximum safety) and diverges as $\Psi_{\text{pol}} \to 0$ (danger). Differentiating along the lifting solution:

$$\dot{\mathcal{V}}(t) = -\frac{d}{dt} \ln \psi_*(t) = -\frac{\dot{\psi}_*(t)}{\psi_*(t)}. \tag{29}$$

Thus $\mathcal{V}$ is non-increasing if and only if $\psi_*(t)$ is non-decreasing, which holds whenever the destination is safer than the start, i.e., $\Psi_{\text{pol}}(D(t), t) \geq \Psi_{\text{pol}}(S(t), t)$.

9.3 Equilibria and Their Stability

A political equilibrium is a point $\mathbf{z}^*$ where $\boldsymbol{\gamma}'(t) = \mathbf{0}$, requiring simultaneously

$$\dot{\psi}_* = \partial_t \Psi_{\text{pol}}(\mathbf{z}^*, t) \qquad \text{and} \qquad \Pi^{\perp}(\mathbf{z}^*, t)\left[D(t) - S(t)\right] = \mathbf{0}.$$

The first condition equates the prescribed safety drift to the autonomous drift at the equilibrium; the second requires $D(t) - S(t)$ to be parallel to $\nabla_{\mathbf{x}} \Psi_{\text{pol}}$ at $\mathbf{z}^*$. Generically these two scalar equations in three unknowns determine a curve of equilibria, whose stability is governed by the Jacobian of (18).

10 Bifurcations and Phase Transitions

10.1 Critical Points of the PDP

A critical point of $\Psi_{\text{pol}}(\cdot, t)$ is a point $\mathbf{x}^*$ where $\nabla_{\mathbf{x}} \Psi_{\text{pol}}(\mathbf{x}^*, t) = \mathbf{0}$. At such a point the safety gradient collapses and the lifting ODE (18) is singular.

Proposition 10.1. *For generic positions of $F_1(t), \ldots, F_m(t)$, the PDP $\Psi_{\text{pol}}(\cdot, t)$ has finitely many critical points in $\mathbb{B}^3$. These are saddle points of Ψ_{pol} located near the Poincaré-geodesic segments connecting pairs of forbidden zones.*

Proof sketch. The gradient $\nabla_{\mathbf{x}} \Psi_{\text{pol}}$ is a smooth vector field on the compact region $\overline{\mathbb{B}^3} \setminus \mathcal{F}(t)$. Its zeros are isolated by the Morse-theoretic structure of Ψ_{pol}: since $\Psi_{\text{pol}} \to 0$ near each F_k and Ψ_{pol} is bounded away from 0 elsewhere, each local minimum lies at an F_k and each saddle lies on the Poincaré geodesic between a pair (F_j, F_k). For generic positions, these saddles are non-degenerate and therefore isolated. □

A *phase transition* occurs at time t^* if the prescribed safety schedule $\psi_*(t)$ achieves the value $\Psi_{\text{pol}}(\mathbf{x}^*(t^*), t^*)$ at a saddle point $\mathbf{x}^*$. At this moment the lifting ODE has a non-isolated solution and the avoidance path becomes ambiguous: this is the three-dimensional counterpart of the critical-value crossing of the planar theory, and corresponds to a historical turning point where multiple qualitatively different political futures become simultaneously available.

10.2 Catastrophe-Theoretic Classification

Near a phase transition, the local normal form of Ψ_{pol} near a critical saddle is classified by the standard hierarchy of catastrophe theory [7, 9, 8].

Table 2: Catastrophe-theoretic classification of political turning points in $\mathbb{B}^3$. New singularity types arising from the third dimension are marked with $\star$.

Singularity	Codim.	Political interpretation
Fold (A_2)	1	A single turning point: an incremental reform path reaches its limit and cannot be continued without reversal.
Cusp (A_3)	2	Two paths merge at the transition; moderate and radical routes become indistinguishable.
Swallowtail (A_4)	3	Three paths merge; a rare event at the intersection of multiple simultaneous political forces.
Elliptic umbilic (D_4^-)*	3	Rotationally symmetric transition in the (x, z)-plane: the Nationalism–Globalism axis creates a circular attractor ring around F_4, threatening centre-left and centre-right equally.
Hyperbolic umbilic (D_4^+)*	3	Saddle-type transition in full 3D: F_4 is approached simultaneously from the authoritarian ($-y$) and nationalist ($-z$) directions, triggering an asymmetric bifurcation.
Butterfly (A_5)*	4	Four paths merge: an exceptional convergence of left–right, auth–lib, and nat–glob crises.

The elliptic umbilic is the genuinely new singularity of the three-dimensional setting. It arises near F_4 when trade disruption and migration stress drive the Ethno-Nationalist Isolationism zone into the central ball along the z-axis, creating a rotationally symmetric danger ring in the (x, z)-plane. Societies approaching this ring from any direction face equivalent danger regardless of whether they originate from the centre-left or centre-right.

10.3 Avoiding Phase Transitions by Design

For each fixed t, the set of critical values of $\Psi_{\text{pol}}(\cdot, t)$ is finite (by Proposition 10.1), so the space of admissible safety schedules is open and dense in the compact-open topology. In practical terms: there always exists a smooth safe policy path with no phase transitions, though it may require more deliberate steering than naive linear interpolation would provide.

11 Numerical Implementation

11.1 Adaptive Runge–Kutta Integration

The three-dimensional Lifting ODE (18) is a system of three scalar first-order ODEs and is solved by an adaptive RK4-5 (Dormand–Prince) scheme. At each step, the algorithm evaluates Ψ_{pol}, $\nabla_{\mathbf{x}} \Psi_{\text{pol}}$, and $\partial_t \Psi_{\text{pol}}$, all in closed form via (7), (14), and (11).

11.2 Condition Number Monitoring

The critical numerical risk is a small gradient denominator. Define the local condition number

$$\kappa_3(t) = \frac{\Psi_{\text{pol}}(\boldsymbol{\gamma}(t), t)}{\|\nabla_{\mathbf{x}} \Psi_{\text{pol}}(\boldsymbol{\gamma}(t), t)\|}. \tag{30}$$

When $\kappa_3(t)$ falls below a prescribed threshold $\kappa_{\min}$, the trajectory is approaching a saddle point of Ψ_{pol}. The algorithm should then apply a small perturbation to $\psi_*(t)$ to steer around the critical value, analogously to the correction procedure of the planar theory.

11.3 Computational Formulae

Table 3: Closed-form expressions for the numerical implementation.

Quantity	Formula	Implementation note
$\rho_k^2(\mathbf{x},t)$	$\|\mathbf{x}-F_k\|^2/(1-2\langle\mathbf{x},\,F_k\rangle+\|\mathbf{x}\|^2\,\|F_k\|^2)$	Use squared form; take sqrt only when needed
$\Psi(\mathbf{x},t)$	$\prod_{k=1}^4 \rho_k(\mathbf{x},t)$	Evaluate in log domain: $\ln\Psi=\sum_k \ln\rho_k$
$\nabla_{\mathbf{x}}\Psi$	$\Psi\sum_k \nabla_{\mathbf{x}}\rho_k/\rho_k$, with $\nabla_{\mathbf{x}}\rho_k$ from (14)	Monitor $\|\nabla\Psi\|$ for saddle proximity
$\partial_t\Psi$	$\Psi\sum_k(\dot N_k D_k - N_k\dot D_k)/(2\rho_k^2 D_k^2)$, with $\dot N_k,\dot D_k$ from (10)	Requires $\dot F_k$ from auxiliary ODE
$\psi_*(t)$	$\Psi(S,t)^{(T-t)/T}\cdot\Psi(D,t)^{t/T}$	Log interpolation for numerical stability
$\boldsymbol{\gamma}'(t)$	Eq. (18): three real components	Evaluate $\Pi^\perp$ as $I-\hat n\hat n^\top$, $\hat n=\nabla\Psi/\|\nabla\Psi\|$

11.4 Real-Time Political Monitoring

The three new observable state variables on the Nationalism–Globalism axis are drawn from publicly available data streams: trade openness $\tau(t)$ (World Bank Trade as % of GDP); migration pressure $M(t)$ (UNHCR annual displacement statistics [13]); and trade disruption $\mathcal{T}(t)$ (WTO global trade disruption index and import tariff shock indices). Combined with the existing variables feeding the x-y dynamics, the algorithm constitutes a real-time volumetric policy navigation system, continuously updating $\boldsymbol{\gamma}(t)$ as the socioeconomic environment evolves.

12 Structural Comparison: Unit Disk versus Unit Ball

Table 4 collects the structural correspondences and contrasts between the two-dimensional and three-dimensional frameworks.

Table 4: The unit disk $\mathbb{D}$ (two dimensions) versus the unit ball $\mathbb{B}^3$ (three dimensions): a systematic comparison.

Feature	$\mathbb{D}$ **(2D)**	$\mathbb{B}^3$ **(3D)**
Safety encoding	Blaschke product $B(z,t)$; holomorphic, degree m	PDP $\Psi(\mathbf{x},t)$; real, C^∞, product form
Forbidden points	Zeros of B, simple	Zeros of Ψ, isolated
Lifting ODE	Complex scalar ODE on $\mathbb{C}$	Real vector ODE on $\mathbb{R}^3$; safety + tracking components
Topology of complement	$\pi_1(\mathbb{D}\setminus\mathcal{F})\cong\mathbb{Z}^m$; non-trivial	$\pi_1(\mathbb{B}^3\setminus\mathcal{F})=\{1\}$; trivial
Topological invariant	Integer winding number $n_k(t)\in\mathbb{Z}$	Real solid angle $\Omega_k(t)\in\mathbb{R}_{\geq 0}$
Monodromy	Integer-valued; $n_k(T)-n_k(0)\in\mathbb{Z}$	Absent for point obstacles
Hysteresis mechanism	Winding-number monodromy	Solid-angle drift (real-valued, direction-sensitive)
Critical points	Zeros of $\partial_z B$; finitely many	Saddle points of Ψ; near geodesics between F_k
Phase transitions	$w(t)$ crosses critical value; fold/cusp/swallowtail	$\boldsymbol{\gamma}(t)$ passes near saddle; adds umbilic and butterfly
Political axes	Left–Right, Auth–Lib	+ Nationalist–Globalist
Forbidden zones	3 (Stalinist, Fascist, Pure Auth.)	4 (+ Ethno-Nat. Isolationism)

13 Conclusion

We have extended the dynamic smooth avoidance framework from the unit disk $\mathbb{D}$ to the unit ball $\mathbb{B}^3$, adding the Nationalism–Globalism axis as the third dimension of ideological space.

Architectural shift. The holomorphic Blaschke product, which has no natural analogue in odd real dimensions, is replaced by the pseudohyperbolic distance product $\Psi(\mathbf{x},t)=\prod_k \rho(\mathbf{x},F_k(t))$, constructed from the real Möbius automorphisms of $\mathbb{B}^3$ (3). All fundamental properties — smoothness (Proposition 4.3), zero-locus encoding (Definition 4.1), time differentiability (11) — are preserved, and the Dynamic Lifting Theorem (Theorem 6.3) extends the existence-uniqueness result to the three-dimensional setting.

Topological shift. The fundamental group $\pi_1(\mathbb{B}^3\setminus\mathcal{F})=\{1\}$ is trivial for point obstacles, eliminating winding-number monodromy (Proposition 7.1, Corollary 7.2). The relevant invariant shifts to the solid angle $\Omega_k(t)$ governed by the transport equation (21). Ideological hysteresis persists as solid-angle drift (Definition 7.5): a real-valued, direction-sensitive measure of how deeply and from which direction a society has approached each dangerous attractor.

Political enrichment. The three-dimensional Political Compass introduces a fourth forbidden zone — the Ethno-Nationalist Isolationism attractor F_4 — driven by trade disruption and migration pressure (25). The safety index $\Phi_3(\mathbf{x},t)$ resolves positions that are indistinguishable on the classical two-dimensional compass. The catastrophe-theoretic classification acquires two genuinely new singularity types (elliptic and hyperbolic umbilic, butterfly) that have no counterpart in the planar theory.

Future extensions include modelling forbidden zones as one-dimensional curves or two-dimensional surfaces in $\mathbb{B}^3$, which would restore non-trivial fundamental group structure. A game-theoretic embedding of multiple simultaneously navigating societies, coupled through shared macroeconomic state variables, would allow modelling geopolitical alliances and competitive ideological pressures in three dimensions.

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Glossary of New Terms (Three-Dimensional Extension)

Unit Ball $\mathbb{B}^3$. The open unit ball $\{\mathbf{x} \in \mathbb{R}^3 : \|\mathbf{x}\| < 1\}$, equipped with the Poincaré metric of real hyperbolic 3-space $\mathbb{H}^3$; the three-dimensional replacement for the unit disk $\mathbb{D}$.

Möbius Automorphism of $\mathbb{B}^3$. The map $\varphi_{\mathbf{a}} : \mathbb{B}^3 \to \mathbb{B}^3$ defined by (3), sending $\mathbf{a} \mapsto \mathbf{0}$ and satisfying $\varphi_{\mathbf{a}} \circ \varphi_{\mathbf{a}} = \mathrm{id}$; the three-dimensional analogue of the disk automorphism $\varphi_a(z) = (z - a)/(1 - \bar{a}z)$.

Pseudohyperbolic Distance ρ. The distance function $\rho(\mathbf{x}, \mathbf{a}) = \|\varphi_{\mathbf{a}}(\mathbf{x})\|$ satisfying (5); takes values in $[0, 1)$, vanishes iff $\mathbf{x} = \mathbf{a}$, and is invariant under the Möbius group of $\mathbb{B}^3$.

Pseudohyperbolic Distance Product (PDP). The function $\Psi(\mathbf{x}, t) = \prod_{k=1}^{m} \rho(\mathbf{x}, F_k(t))$; the three-dimensional structural replacement for the modulus $|B(z, t)|$ of the time-varying Blaschke product. It vanishes precisely on $\mathcal{F}(t)$ and lies in $(0, 1)$ elsewhere.

Nationalism–Globalism Axis. The third coordinate axis $z \in (-1, +1)$ in the extended Political Compass, with $z = -1$ representing extreme nationalism and $z = +1$ representing extreme globalism; empirically orthogonal to the classical Left–Right (x) and Authoritarian–Libertarian (y) axes.

Ethno-Nationalist Isolationism Zone (F_4). The fourth dynamic forbidden zone, located in the Auth-Right–Nationalist octant of $\mathbb{B}^3$, driven by trade disruption and migration pressure; absent from the two-dimensional model and the principal new danger introduced by the third axis.

Solid Angle $\Omega_k(t)$. The real-valued three-dimensional invariant replacing the integer winding number $n_k(t)$ of the planar theory; defined by (19) and governed by the transport equation (21). It measures the total spherical arc swept by the radial direction from the agent to F_k.

Solid-Angle Hysteresis. The persistence of non-zero net solid angle when a society returns to its original coordinates in $\mathbb{B}^3$; the three-dimensional replacement for winding-number monodromy, richer because it is real-valued and direction-sensitive rather than integer-valued.

Elliptic Umbilic (D_4^-). A new singularity type (codimension 3) arising in the three-dimensional catastrophe classification near F_4; corresponds to a rotationally symmetric danger ring in the (x, z)-plane affecting centre-left and centre-right societies equally.

Three-Dimensional Political Safety Index Φ_3. The magnitude $\Phi_3(\mathbf{x}, t) = \Psi_{\mathrm{pol}}(\mathbf{x}, t) \in [0, 1]$; a real-time risk indicator that distinguishes ideological positions indistinguishable on the two-dimensional compass by incorporating the Nationalism–Globalism axis.

The End